

TOPIC NAME: LSTM

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Long Short-Term Memory (LSTM)

We use LSTM instead of a basic (vanilla) RNN.

Problems with Vanilla RNN: Vanishing & Exploding Gradient problem.

During backpropagation through time, gradients:

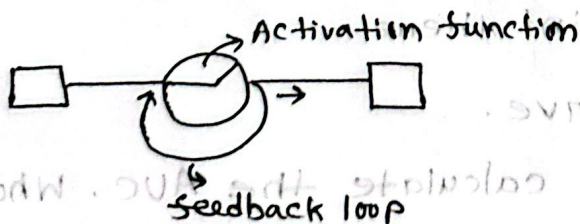
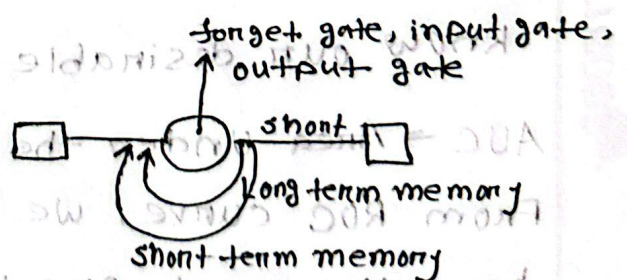
i) Shrink to near zero (vanish) $\rightarrow$ model forgets early informationii) Grow too large (explode) $\rightarrow$ unstable training

As a result, vanilla RNNs:

i) Remember only short term context

ii) Perform poorly on long sequences (text, speech, time series)

So we use LSTM as it remembers important information for a long time and trains stably.

RNNLSTM

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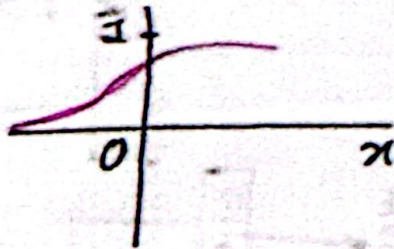
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Activation Functions :

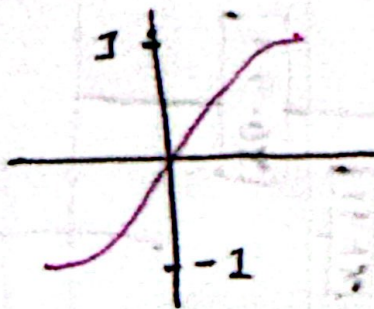
Sigmoid :

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

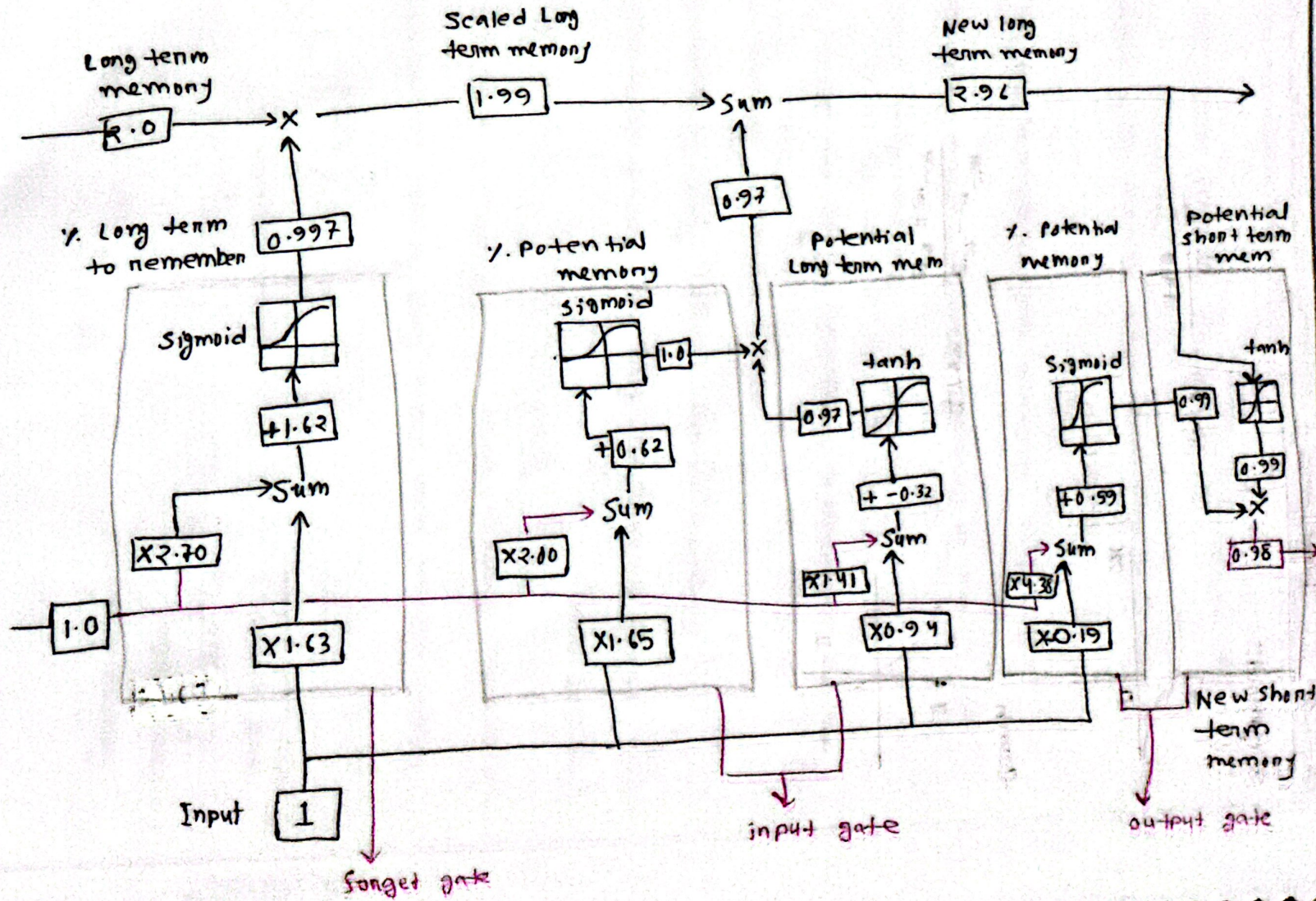


tanh :

$$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



LSTM structure



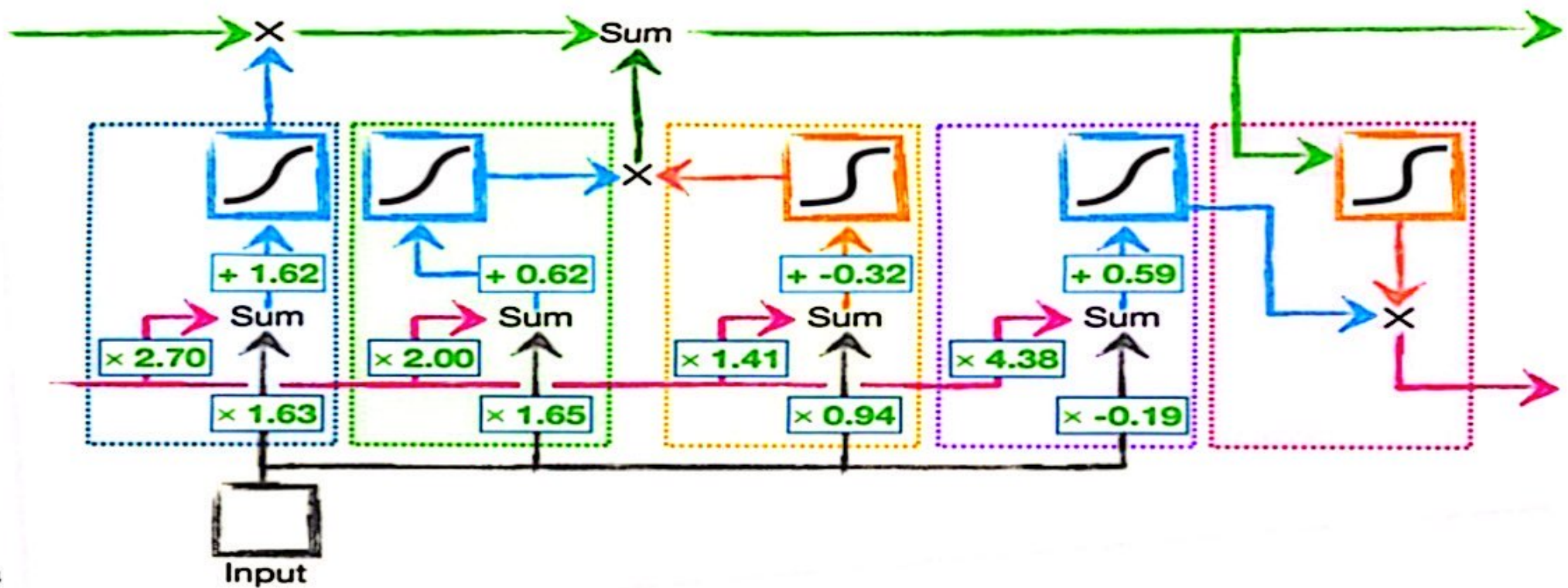
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7. If the input for day1 and day2 are 1.0 and 0.5 respectively in the following LSTM cells, what will be the output for day3?



TOPIC NAME: Assignment 2

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⑦ Input for day 1 = 1.0

Input for day 2 = 0.5

To find output for Day 3, we need to perform

a forward pass through LSTM cell for three

~~consecutive time steps (t = 1, 2, 3)~~

Day 1 (x₁ = 1.0)

$$f_1: \sigma(1.0 \times 1.63 + 0 \times 2.70 + 1.62) = \sigma(3.25) \approx 0.9627$$

$$i_1: \sigma(1.0 \times 1.65 + 0 \times 2.00 + 0.62) = \sigma(2.27) \approx 0.9064$$

$$i_2: \tanh(1.0 \times 0.94 + 0 \times 1.41 - 0.32) = \tanh(0.62) \approx 0.5511$$

$$o_1: \sigma(1.0 \times (-0.19) + 0 \times 4.38 + 0.59) = \sigma(0.40) \approx 0.5987$$

$$\text{Cell state } c_1: (0.9627 \times 0) + (0.9064 \times 0.5511) = 0.4995$$

$$\text{New short-term memory: } 0.5987 \times \tanh(0.4995) \approx 0.2764$$

Day 2 (x₂ = 0.5)

$$f_2: \sigma(0.5 \times 1.63 + 0.2764 \times 2.70 + 1.62) = \sigma(3.181) \approx 0.9599$$

$$i_1': \sigma(0.5 \times 1.65 + 0.2764 \times 2.00 + 0.62) = \sigma(1.998) \approx 0.8806$$

$$i_2': \tanh(0.5 \times 0.94 + 0.2764 \times 1.41 - 0.32) = \tanh(0.54) \approx 0.4930$$

$$o_2: \sigma(0.5 \times (-0.19) + 0.2764 \times 4.38 + 0.59) = \sigma(1.706) \approx 0.8463$$

$$c_2: (0.9599 \times 0.4995) + (0.8806 \times 0.4930) = 0.9136$$

GOOD LUCK™

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§ term DAY A

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New short term memory: $0.8463 \times \tanh(0.9136)$

≈ 0.6117 (Ans)